

P1 exam: Days 1-6

Quadratics, functions, coordinate geometry and circles

YOUR AIM TODAY

Complete a 60-mark, 75-minute closed-book exam and show every essential step.

YOUR RULE

No notes and no answer sheet. A calculator is allowed. Give exact answers unless a decimal is requested. If stuck, write a relevant formula and continue.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

Before you begin

This paper is designed to show what to repair next, not merely produce a score.

Item	Requirement
Time	75 minutes
Total	60 marks
Topics	Quadratics; discriminant and inequalities; functions; straight lines; circles
Method	Show algebra, substitutions and equations. Unsupported answers may lose method marks.
Finish	Check signs, brackets, exact values and whether every part was attempted.

TEACHER SETUP

Print only this question paper for Rohan. Keep the separate mark scheme closed until the full 75 minutes have ended.

Student declaration

I completed this paper without notes or answers.

Signature: _____ Start: _____ Finish: _____

Score record

Quadratics /24	Functions /10	Coordinates /12	Circles /14	Total /60

Quadratics

Questions 1-3 test algebra, roots, graphs and parameter reasoning.

QUESTION 1

Core quadratic skills

[8 marks]

(a) Expand and simplify $(2x - 3)(x + 4)$. [2]

(b) Complete the square: $x^2 - 8x + 11$. [2]

(c) Hence state the coordinates of the turning point of $y = x^2 - 8x + 11$. [1]

(d) Solve $x^2 - 8x + 11 = 0$, giving exact answers. [3]

QUESTION 2

Roots and the discriminant

[8 marks]

(a) Solve $3x^2 - 7x - 6 = 0$. [3]

(b) State the discriminant of $ax^2 + bx + c$ and explain what a zero discriminant means. [2]

(c) Find k if $x^2 + 6x + k = 0$ has a repeated root. [3]

Quadratic inequalities and parameters

Use roots and the shape of the parabola.

QUESTION 3

Inequalities and a parameter

[8 marks]

(a) Solve $x^2 - x - 12 < 0$. [3]

(b) Solve $2x^2 + 5x - 3 \geq 0$. [3]

(c) Find the set of values of p for which $x^2 + px + 9 = 0$ has no real roots. [2]

CHECKPOINT

You have completed 24 marks. Suggested elapsed time: 30 minutes.

Functions

Read composites from the inside and reverse operations for inverses.

QUESTION 4

Functions, composites and inverses

[10 marks]

GIVEN

$f(x)=2x-5$ and $g(x)=x^2+1$, for real x .

(a) Find $f(-3)$. [1]

(b) Find $fg(x)$, simplifying your answer. [2]

(c) Find $gf(x)$, simplifying your answer. [2]

(d) Solve $fg(x)=15$. [2]

(e) Find f inverse of x . [2]

(f) Explain why g does not have an inverse function when its domain is all real numbers. [1]

Coordinate geometry

Build equations from gradients and points.

QUESTION 5

Straight lines and perpendicular geometry

[12 marks]

GIVEN

$A=(2,-1)$ and $B=(8,7)$.

(a) Find the gradient of AB. [2]

(b) Find the midpoint of AB. [1]

(c) Find the exact length AB. [2]

(d) Find an equation of line AB in the form $y=mx+c$. [3]

(e) Find an equation of the perpendicular bisector of AB. [4]

Circles

Use completed-square form and radius-tangent geometry.

QUESTION 6**Circle equations and a tangent**

[14 marks]

GIVEN

Circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$.

(a) Write the equation in the form $(x-a)^2 + (y-b)^2 = r^2$. [3]

(b) State the centre and radius of C. [2]

(c) Show that $P=(6,2)$ lies on C. [2]

(d) Find the gradient of the radius from the centre to P. [2]

(e) Hence find the equation of the tangent to C at P. [3]

(f) Find the two points where C meets the line $y=-2$. [2]

END OF PAPER

Stop after 75 minutes. Check every page, then send the complete paper for marking before viewing the mark scheme.